

EaaS — Product Requirements Document

Index & Glossary

Document Structure

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eaas-prd/
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├── README.md                ← You are here
├── PRD.md                   ← Main Product Requirements Document
├── NON_FUNCTIONAL_REQUIREMENTS.md ← Performance, security, SLA specs
|
├── user-stories/
|   └── USER_STORIES.md     ← All epics, user stories & acceptance
criteria
|
├── diagrams/
|   ├── ERD.mermaid         ← Entity Relationship Diagram (all
tables)
|   ├── STATE_MACHINE.mermaid ← Escrow status state machine
|   ├── SEQUENCE_HAPPY_PATH.mermaid ← Happy path sequence diagram
|   ├── SEQUENCE_DISPUTE_FLOW.mermaid ← Dispute resolution sequence diagram
|   └── schema.sql          ← Full PostgreSQL database schema
|
├── architecture/
|   └── ARCHITECTURE.md      ← HLA, LLA, microservices, K8s, event
map
|
└── api-specs/
    └── openapi.yaml        ← Full OpenAPI 3.0 specification
```

How to View Diagrams

The `.mermaid` files can be rendered at:

- <https://mermaid.live> — paste file contents
- **VS Code** — install "Mermaid Preview" extension
- **GitHub** — renders Mermaid in markdown automatically

The `openapi.yaml` can be viewed at:

- <https://editor.swagger.io> — paste file contents

Glossary

Term	Definition
EaaS	Escrow as a Service — the product name
Escrow	A financial arrangement where a neutral third party holds funds until conditions are met
Merchant	The seller / business receiving payment through EaaS
Customer	The buyer making payment through EaaS
Partner	A third-party platform integrating EaaS via API
PSP	Payment Service Provider (Paystack in v1)
KYC	Know Your Customer — identity verification process
AML	Anti-Money Laundering — regulatory compliance requirement
BVN	Bank Verification Number — Nigerian bank identity number
CAC	Corporate Affairs Commission — Nigerian business registration body
CBN	Central Bank of Nigeria — financial regulator
Idempotency Key	A unique key preventing duplicate processing of the same request
State Machine	A formal model of an escrow transaction's lifecycle states and transitions
HPA	Horizontal Pod Autoscaler — Kubernetes auto-scaling mechanism
DLQ	Dead Letter Queue — RabbitMQ queue for failed messages
HMAC	Hash-based Message Authentication Code — used to verify webhook authenticity
Confirmation Window	72-hour period after delivery during which customer can confirm or dispute
Auto-Release	Automatic fund release to merchant after 72-hour confirmation window expires
Payout	Transfer of escrow funds to merchant's bank account
Refund	Return of escrow funds to customer's payment method

Term	Definition
Dispute	A formal claim by the customer that the product was not received or not as described
Arbitration	Admin review and resolution of a dispute
Escrow Fee	EaaS platform fee charged per transaction (default 1.5%)
Merchant Amount	Amount merchant receives = escrow amount minus escrow fee
Total Charge	Amount customer pays = product amount plus escrow fee
Reference	Unique escrow transaction identifier (format: TXN-YYYY-XXXXXXXXXX)
Webhook	HTTP callback sent to merchant's URL on escrow state changes
Sandbox	Test environment for merchant integration (no real money)

Key Business Rules Summary

1. **Merchant must be KYC-verified** before receiving any payouts.
2. **Idempotency key is required** on all escrow creation requests.
3. **Customer has 72 hours** after delivery to confirm or dispute.
4. **Funds auto-release to merchant** if customer does not respond in 72 hours.
5. **Disputes freeze payout** — no auto-release while a dispute is open.
6. **Escrow fee is non-refundable** unless EaaS is at fault.
7. **Minimum transaction amount** is ₦1,000.
8. **Only one active dispute** per escrow transaction at a time.
9. **Dispute must be raised before** the 72-hour window expires.
10. **Admin resolution is final** and binding for both parties.
11. **Cancelled escrows** (unfunded for 24 hours) incur no fee.
12. **Merchant receives net amount** (after fee deduction) — not the gross.

Stakeholder Map

Stakeholder	Role	Key Interest
Product Manager	Owns PRD	Feature scope, user value
Engineering Lead	Technical design	Feasibility, architecture

Stakeholder	Role	Key Interest
Backend Engineers	Implementation	API design, data model
Frontend Engineers	Dashboard & customer UI	API contracts
DevOps Engineer	Deployment	K8s, CI/CD, monitoring
Legal / Compliance	Regulatory	KYC, AML, CBN rules
Finance Team	Reconciliation	Payout accuracy, fees
Customer Support	Dispute handling	Admin tools, escalation
Merchants (external)	Revenue	Reliable payouts, low fees
Customers (external)	Purchase protection	Trust, refund speed



Document History

Version	Date	Author	Changes
1.0.0	June 2026	EaaS Product Team	Initial PRD